Data Cleaning Change Log

DecodeLabs Data Analytics - Project 1: Data Cleaning & Preparation

Dataset	E-Commerce orders, Jan 2023 - Jun 2025
Raw file	Dataset_for_Data_Analytics.xlsx (sheet: Sheet1) - 1,200 rows x 14 columns
Cleaned file	E-Commerce_Sales_Analysis_internship.xlsx (sheet: Data, Excel Table 'Orders') - 1,200 rows x 17 columns
Tool	Microsoft Excel

1. Change Register

Change ID	Description	Impact	Status
CR001	Missing CouponCode: 309 blank cells replaced with the label 'NoCoupon' (categorical field, so no mean/median applies).	Preserved 309 records (25.8% of the data). Coupon column now has 4 explicit values, 0 blanks.	Resolved
CR002	TotalPrice: 29 values carried floating-point artifacts beyond 2 decimals; rounded to 2 decimals.	All 1,200 TotalPrice values now equal Quantity x UnitPrice at 2-decimal precision. Largest adjustment was about 1e-13 (no business impact).	Resolved
CR003	Number display format: UnitPrice and TotalPrice set to currency with 2 decimals.	Consistent numeric precision across all price columns.	Resolved
CR004	Calculated columns added: Year (YEAR of Date), Month (yyyy-mm text), NetRevenue (TotalPrice if status is Shipped or Delivered, else 0).	3 new columns, formula-driven on all 1,200 rows. Supports the Analysis, Pivot and Dashboard tabs.	Resolved
CR005	Data converted to an Excel Table named 'Orders' (A1:Q1201).	Structured references and slicer support. No row changes.	Resolved
CR006	Year and Month columns were displaying with a currency format (e.g. $2,023.00). Format reset to General (cell format and table column default).	Display only. Values were already correct. Year now shows 2023 and Month shows 2023-01.	Resolved

2. Checks Performed With No Change Needed

Check	Finding	Action
Duplicate rows	0 fully duplicated rows	None
Duplicate OrderIDs	0 duplicates; 1,200 unique IDs (all match ORD + 6 digits)	None
Duplicate TrackingNumbers	0 duplicates (all match TRK + 8 digits)	None
Date format	All 1,200 dates already stored as true Excel dates; range 2023-01-01 to 2025-06-30. Formatted yyyy-mm-dd (ISO 8601).	None
Text consistency	No leading/trailing spaces; no case variants. Product 7, PaymentMethod 5, OrderStatus 5, ReferralSource 5 distinct values.	None
Nulls in other columns	0 nulls in every column except CouponCode (see CR001)	None
Numeric validity	Quantity 1-5, ItemsInCart 1-10 (never below Quantity), UnitPrice 11.39-699.93, no negatives, no text-stored numbers	None
Outliers	10 orders have a UnitPrice under $20. Reviewed and left as is: values are plausible and not clearly errors.	None

3. Verification Gate (Project 2 Threshold)

Requirement	Result	Evidence
0% error rate on unique identifiers	PASS - 0 duplicate IDs	1,200 rows, 1,200 distinct OrderIDs
0% error rate on date formats	PASS - 0 incorrect dates	1,200 of 1,200 true date values, all yyyy-mm-dd
0 missing values	PASS	0 blank cells across 17 columns after CR001

4. Notes and Assumptions

• 'NoCoupon' is a label for orders placed without a coupon. It was assumed that a blank means no coupon was used, which the data cannot confirm.

• Coupon discounts are not reflected in TotalPrice in the source data, so coupon analysis compares order size only.

• 487 Cancelled or Pending orders still carry a TrackingNumber. This came from the source and was left unchanged.

• No rows were deleted. Row count is 1,200 before and after cleaning.